ABIN ACHARYA

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EDUCATION

University of Colorado, Colorado Springs (UCCS) *Colorado Springs, CO*
Ph.D. in Computer Science (M.S. earned en route) | GPA: 3.85/4.0 *Aug 2025 – Present*
Focus: AI, Machine Learning, Deep Learning, NLP, Reinforcement Learning, Explainable AI
Relevant Coursework: Artificial Neural Networks, Reinforcement Learning, Operating Systems, Database Designs

Institute of Engineering (IOE), Tribhuvan University *Kathmandu, Nepal*
Bachelor of Computer Engineering | GPA: 3.83/4.0 | Full Merit Scholarship *Aug 2018 – Aug 2023*

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (SQLite, MySQL), JavaScript, HTML/CSS
ML/AI Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, SHAP, LIME
Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Power BI
Tools & Platforms: Git/GitHub, Docker, Linux/Unix, Google Colab, Jupyter, LaTeX, VS Code, Anaconda
Web Development: Django, PHP, JavaScript, RESTful APIs, SQLite, MySQL

TEACHING & PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant *UCCS*
Department of Computer Science *Aug 2025 – Present*

- Support 150+ students across 3 courses (Intro to Software Engineering, Advanced OOP with C#/.NET, Graduate Advanced Software Engineering) through office hours, grading, and lab facilitation.
- Achieved positive feedback rating from student evaluations by providing clear, structured mentorship and responsive academic support.

IT Assistant *Kathmandu, Nepal*
Gurukul Secondary School *Sep 2023 – May 2025*

- Administered school-wide IT infrastructure, including network maintenance, device management, and software updates.
- Provided first-line technical support to staff members and implemented security protocols.

PROJECTS

Explainable AI for Malware Prediction | Python, PyTorch, TabTransformer, CNN, SHAP, LIME, BoolXAI

- Engineered malware detection pipeline using a CNN baseline and TabTransformer on 10K+ samples from the Microsoft Malware Kaggle dataset.
- Integrated BoolXAI for rule-based interpretability, generating human-readable Boolean explanations that improved model transparency without sacrificing accuracy.
- Applied SHAP and LIME to analyze per-feature contributions, identifying top 5 predictive features and enabling trustworthy ML decision-making.

Adaptive RL for Skill-Level Intelligent Tutoring for Algebra Education | Python, DDQN, OpenAI Gym, Streamlit

- Built a reinforcement learning-based intelligent tutoring system using Double DQN to adaptively select algebra problems based on real-time student mastery across 10 skill categories.
- Designed a custom RL environment with reward shaping and state representation encoding student performance history, achieving 85%+ simulated mastery convergence.
- Developed Streamlit demo interface for interactive tutoring sessions with LLM-generated hints.

Virtual Try-On Using Neural Networks | Python, PyTorch, GANs, U-Net, OpenCV

- Developed a GAN-based virtual try-on system using Geometric Matching Module and U-Net Try-On Module with TPS cloth warping and pose estimation.

- Achieved 8% Normalized Mean Squared Error on a 2,000-image test set with sub-second inference time, demonstrating production-viable performance.

Discussion Forum Web Application | Python, Django, HTML/CSS, JavaScript, SQLite

- Built a full-stack Q&A platform with user authentication, content posting, search, and trending post algorithms serving 50+ concurrent test users.
- Implemented content-based recommendation engine using cosine similarity, increasing average session engagement by 25% in user testing.

HONORS & AWARDS

- Fully Funded Ph.D. Scholarship, University of Colorado, Colorado Springs (UCCS), 2025 – Present
- Full Undergraduate Merit Scholarship, Institute of Engineering (IOE), 2018 – 2023
- Mahatma Gandhi Scholarship, Indian Embassy, 2016 – 2018
- Licensed Professional Engineer, Nepal Engineering Council (NEC)